

EESSI

EUROPEAN ENVIRONMENT FOR SCIENTIFIC SOFTWARE INSTALLATIONS

2 July 2026

https://github.com/EESSI/meetings/wiki

This meeting is being recorded

- Be aware: If you speak up, you will be in the recording
- Feel free to raise questions via Zoom chat instead
- We'll do informal Q&A at the end (after stopping recording)
- Recording will be made available via EESSI YouTube channel:

  https://www.youtube.com/@eessi_community

Agenda



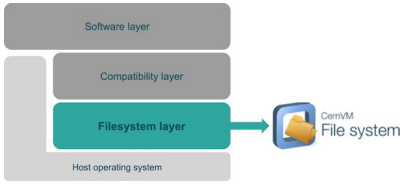
- Quick introduction by new people
- Progress update per EESSI layer
- Update on EESSI production repository `software.eessi.io` (+ RISC-V)
- Update on EESSI test suite, build-and-deploy bot, documentation, etc.
- Status update of EESSI on EuroHPC systems
- Progress on AMD GPU support
- Using EESSI as a base software stack
- AWS sponsorship update
- EESSI governance
- Upcoming/recent events
- Q&A

Quick introduction by new people

New people on the call: feel free to introduce yourself!

- Who are you, where do you work, on what?
- Why are you interested in the EESSI project?
- Are you planning to actively contribute,
  and if so, to which aspect(s) of the project?

Progress update: filesystem layer



- No issues with the CernVM-FS infrastructure, all servers current have:
  - Latest kernel version
  - CernVM-FS version 2.13.3
- New release to provide updated `cvmfs-config-eessi` packages to configure CernVM-FS
  - https://github.com/EESSI/filesystem-layer/releases/tag/v0.6.0
  - Includes recently added Stratum-1 mirror server hosted by RAL (UK)
  - Includes new options, e.g. to use a CDN (http://s1eessieu1-cvmfs.openhtc.io, etc.)
  - Configuration files for EESSI pilot (eessi-hpc.org) have been dropped
- Additional (internal) Stratum 0 has been set up for testing purposes

[Bob, Kenneth]

Progress update: filesystem layer monitoring



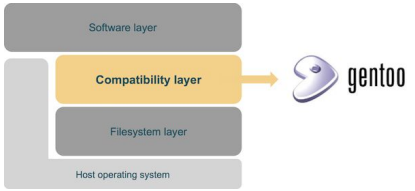
Size is (still) growing with ~1GB/day

Currently ~1TB of disk space is required
for a full Stratum-1 mirror of all EESSI repositories

This information is now presented at https://status.eessi.io/trends.html

[Bob, Kenneth]

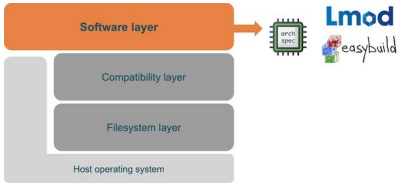
Progress update: compatibility layer



- Compatibility layer for EESSI 2026.06 is ingested
  - https://github.com/EESSI/compatibility-layer/pull/235
  - Updated versions of packages, e.g. glibc 2.43, Lmod 9.2.4
  - Use RPATH for compatibility layer binaries
  - All Portage configuration files are now created by Ansible -> gentoo-overlay repository only used for ebuild files
  - Clearer split between configuration of different EESSI versions
  - Many variant symlinks, with default targets in `/cvmfs/software.eessi.io/defaults/compat`

[Bob, Kenneth]

Progress update: software layer (1/2)



Highlights of recently merged software PRs (2023.06):

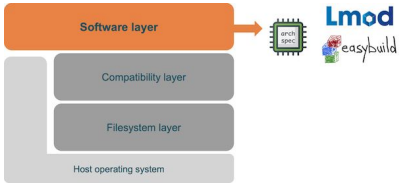
- 655 distinct software packages available now,
  4 merged PRs since last update meeting (7 May'26)

Highlights of recently merged software PRs (2025.06):

- 484 distinct software packages now available,
  29 merged PRs since last update meeting (7 May'26)
- More top level applications available:
  GROMACS 2025.4 w/ CUDA 12.9.1 (PR #1482) & 2026.2 w/ CUDA 12.9.1 (PR #1524), Qt6 6.9.3 (PR #1486), GROMACS 2026.2 (PR #1497), scikit-bio 0.7.2 (PR #1494), QuantumESPRESSO 7.5 (PR #1502), Siesta 5.4.2 (PR #1503), ESPResSo 4.2.2 (PR #1513) & 5.0.1 (PR #1517), Nextflow 26.04.0 (PR #1514), OpenFOAM 13 (PR #1515), LAMMPS Jul2025 update 4 (PR #1539)
- ROCm (see later slide) (PR #1473 & PR #1499)

[Kenneth, Thomas, Richard, Pedro, Bob, Lara, Caspar, Satish, Alan, Mattias, Danilo, Helena …]

Progress update: software layer (2/2)

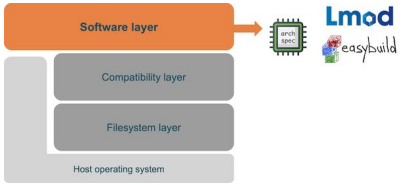


Rebuilds of existing software installations:

- EESSI 2025.06
  - Rebuild `Siesta/5.4.2-foss-2024a` (PR #1534)
  - Rebuild `Rust/1.91.1-GCCcore-14.3.0` to fix RPATH linking (PR #1533)

[Kenneth, Thomas, Richard, Pedro, Bob, Lara, Caspar, Satish, Alan, Helena …]

Progress update: software-layer-scripts



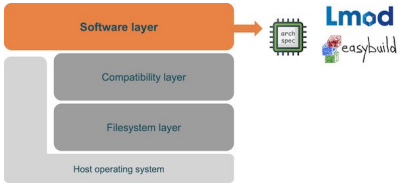
Highlights of recently merged software-layer-scripts PRs:

- Several PRs for AMD GPU support (PR #228, PR #255, PR #256, PR #257, PR #259)
- add Zen5 support to archdetect (PR #162)
- Add microarchitecture detection for RISC-V clusters at HCA (PR #122)
- Allow building for a list of accelerators in a single build (PR #210)

+ various other minor PRs: #25, #99, #162, #222, #225, #226, #227, #232, #233, #239, #240, #242, #244, #246, #248, #250, #251, #252, #254, #258, #260

[Kenneth, Alan, Bob, Richard, Thomas, Aayush …]

Progress update: RISC-V



RISC-V software stack being built via https://github.com/EESSI/dev.eessi.io-riscv

- See page in documentation: https://www.eessi.io/docs/repositories/dev.eessi.io-riscv/
- Available software page 🎉 https://www.eessi.io/docs/available_software_riscv/
- Installations targeting `/cvmfs/dev.eessi.io/riscv` repo
  - Same installation workflow through bot
  - Built on top of EESSI 2025.06 compatibility layer for RISC-V (available in `software.eessi.io`)
- >300 installations have already been ingested for `riscv64/generic` 🎉
  - `foss/2025a`, `foss/2025b`, and **`lfoss/2025b`** are available
- New CPU target has been added: `riscv64/sifive/p550`
  - `foss/2025b` is already available
- Planning to also look into:
  - More end-user applications (GROMACS 2026.02 recently added)
  - Going beyond `riscv64/generic` as target, also optimize for Banana Pi (RISC-V CPU w/ vector instructions)

[Bob, Julian, Kenneth, …]

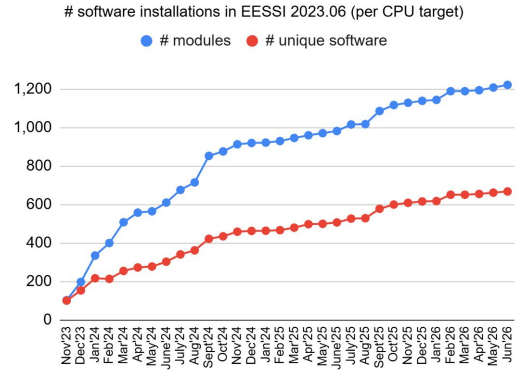
EESSI production repository

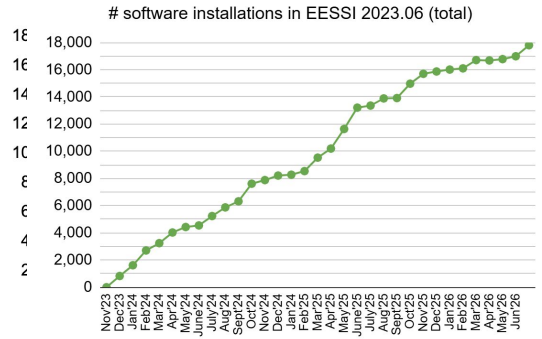
eessi.io/docs

Status update on **production-ready EESSI repository** `software.eessi.io`

Version **2023.06** is being populated with software via PRs to software-layer repo + build-and-deploy bot

- 14 fully supported CPU targets: see eessi.io/docs/software_layer/cpu_targets
- Support for NVIDIA GPUs is in place, see eessi.io/docs/gpu
- **Currently: 1,222 software installations per CPU target**
  (+21 in May+Jun'26)
  - **668 different software projects** (+13 in May+Jun'26)
    17,800 software installations (across 14 CPU targets, +1021 in May+Jun'26)
- Current efforts:
  - Adding more GPU (CUDA) software installations
  - Processing incoming contributions
- See overview of available software: https://eessi.io/docs/available_software
- **Attention is shifting to EESSI version 2025.06 (see next slide)**





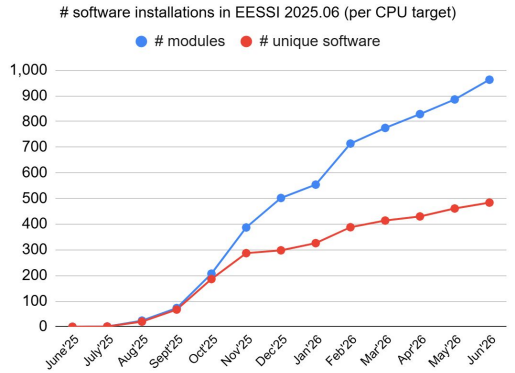
EESSI production repository

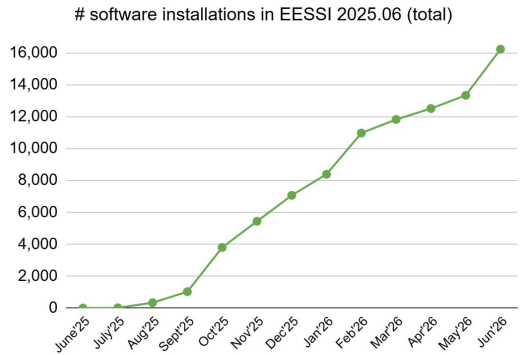
eessi.io/docs

Status update on **production-ready EESSI repository** `software.eessi.io`

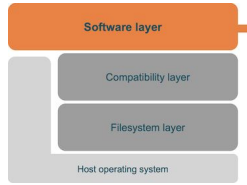
Version **2025.06** is being populated with software via PRs to software-layer repo + build-and-deploy bot

- **15** fully supported CPU targets: see eessi.io/docs/software_layer/cpu_targets
- Support for NVIDIA GPUs is in place, see eessi.io/docs/gpu
  - AMD GPU support is there, but not exposed by archdetect
- **Currently: 963 software installations per CPU target** (+134 in May+Jun'26)
  - **484 different software projects** (+54 in May+Jun'26)
    16,220 software installations (across 15 CPU targets, +3715 in May+Jun'26)
- See overview of available software: https://eessi.io/docs/available_software
- Current focus:
  - Add CUDA installations
  - Adding more software, processing incoming contributions





Bot for building + deploying software





https://github.com/EESSI/eessi-bot-software-layer + https://github.com/EESSI/filesystem-layer

Current bot version 0.12.0 has been released (10 June'26) release notes

Developments since then

- [MERGED] work towards GitLab support, `bot:help` command (PR #376)
- [UNDER REVIEW] improved test coverage (PR #379)
- adding instructions to set up bot instance for GitLab repositories (ongoing work, no PR yet)

Aiming for merge of branch gitlab into develop and then a new release so that those interested in GitLab can start testing it. Note, only the `bot:help` command is currently supported.

[Bob, Caspar, Kenneth, Lara, Pedro, Sam, Sondre, Thomas]

EESSI documentation

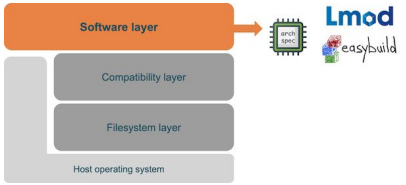
eessi.io/docs - GitHub repo github.com/EESSI/docs

Highlights:

- EESSI Events & Training
  - Added EESSI Webinar Series (+ added links to recordings)
  - Added EESSI Happy Hour (+ added links to recordings)
- Blog posts (https://www.eessi.io/docs/blog/)
  - `lfoss/2025b` now available in the EESSI RISC-V stack
  - MPI at Warp Speed: EESSI Meets Slingshot-11(bis)
- Extensive documentation updates for using EESSI as a base for a local software stack (WIP)
- Plus other minor changes to the documentation (software overview, corrections,...)

[Lara, Pedro, Helena, Bob, Kenneth, Caspar…]

EESSI test suite



eessi.io/docs/test-suite - github.com/EESSI/test-suite

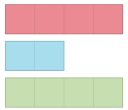
ReFrame

Highlights

- Support for testing multiple EESSI versions in a single run through ReFrame programming environments PR #326
  - Before: `module load EESSI/<version>; reframe --run -n GROMACS`
    - Discovers and runs gromacs tests on GROMACS modules found in `EESSI/<version>`
  - After: `reframe --run`
    - Discovers and runs gromacs tests on GROMACS modules in *any* version
- New release coming soon (aim: 1-2 weeks)

[Caspar, Sam, Satish, Kenneth, Lara]

EasyBuild v5.3.1





Released on 19 June 2026, see https://pypi.org/project/easybuild

Cherrypicked for EESSI relevance:

- enable hwloc, cuFFTMp, and HeFFTe support in GROMACS easyblock
- adapt sanity checks in OpenMPI easyblock for AMD ROCm
- enhance custom easyblock for OpenMPI to add ROCm support
- add easyconfigs for ROCm-based (sub)toolchains: `rocm-compilers/19.0.0-ROCm-6.4.1`, `rompi/2025a`

[Kenneth]

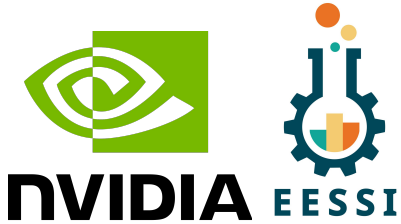
Status update of EESSI on EuroHPC systems

- EESSI is currently available on 8 out 9 current EuroHPC supercomputers:
  LUMI*, Vega, Karolina, Deucalion, MareNostrum 5, Discoverer, MeluXina, Leonardo*
  - Groundwork to make EESSI available on JUPITER is done (not yet in production)
  - Arrhenius is WIP
- EESSI is expected to be available on all 9 (+ upcoming) EuroHPC supercomputers soon, thanks to EuroHPC Federation Platform (EFP)
  - Upcoming EuroHPC AI Factories will also be required to integrate with EFP
  - See also https://docs.my-eurohpc.eu/software-catalog/system-specific/about
- 1st release of EFP happened in April
  - Includes EESSI as base of Federated Software Catalog component (in beta)
  - More info on EFP: https://my-eurohpc.eu

(*) only available on-demand

[Kenneth, Alan, Lara]

Improved NVIDIA GPU support

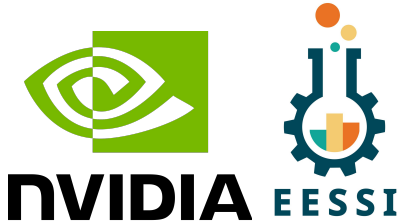


Software-layer:

- Added additional CUDA packages
  - SPH-EXA w/ CUDA 12.4.0 (PR #1452)
  - GROMACS 2025.04 w/ CUDA 12.9.1 (PR #1382)
  - GROMACS 2026.2 w/ CUDA 12.9.1. (PR #1524)
  - hwloc-CUDA (PR #1520)

[Caspar, Bob]

Improved NVIDIA GPU support



Software-layer-scripts

- Allow building for multiple targets in one build job (PR #210)
- Allow CUDA minor version compatibility in LMOD hook checking driver version (PR #226)
- Use newer EasyBuild to install CUDA toolkit in host injections (PR #251)

[Caspar, Bob]

AMD GPU support



EasyBuild

- Higher level ROCm-based toolchains (rompi, rfoss, …) defined at easybuild-framework level
- Easyconfigs for rompi, rfoss, etc: still a work in progress.

[Caspar, Bob]

AMD GPU support



EESSI

- Extensive successful tests with ROCm-enabled OSU & GROMACS installed with EESSI-extend @ LUMI
  - Single GPU: performance on par with native
  - Single node, multi-GPU: OSU performance poor for intermediate message sizes, on par for small/large messages. GROMACS virtually on par.
  - Multi node, multi-GPU: poor performance with EESSI's OpenMPI. Compiled MPICH + host-injected Cray MPICH brings performance on par. Probably mostly a Slingshot issue, not a AMD GPU specific issue.
- Details, see EESSI happy hour - 15 june 2026

Software-layer-scripts:

- Add AMD GPU support to EESSI-extend (PR #206, PR #218)
- Modify architect to allow overriding AMD GPU architectures (PR #215)
- Update EasyBuild hook that checks if GPU software is installed in GPU-prefixes (PR #217)

[Caspar, Bob]

AMD GPU support

Software-layer-scripts:

- Limit ROCm-LLVM builds to 12 cores (PR #224)
- Add GPU tag to builds that depend (via toolchain) on ROCm-LLVM (PR #228)
- Add `rocm-compilers` and `rompi` to list of supported toolchains for `EESSI/2025.06` (PR #255, PR #256, PR #257, PR #259)

Software-layer:

- ROCm-LLVM 19.0.0 w/ ROCm 6.4.1 (PR #1473)
- Rocm-compilers 19.0.0 w/ ROCm 6.4.1 (PR #1499)
- Building towards HIP and RCCL (WIP, PR #1526)

[Caspar, Bob]

Using EESSI as a base for a central software stack



Improved support for using EESSI as a base for a central software stack

- Webinar 1 June 2026 [recording & slides]

Many changes leading up to the webinar to improve support

- Allow sites to define a custom install prefix (PR #232)
- Adapt build scripts from `software-layer-scripts`to improve building-on-top support (PR #239)
- Respect custom install prefix for accelerator installations in `eb_hooks.py`(PR #244)

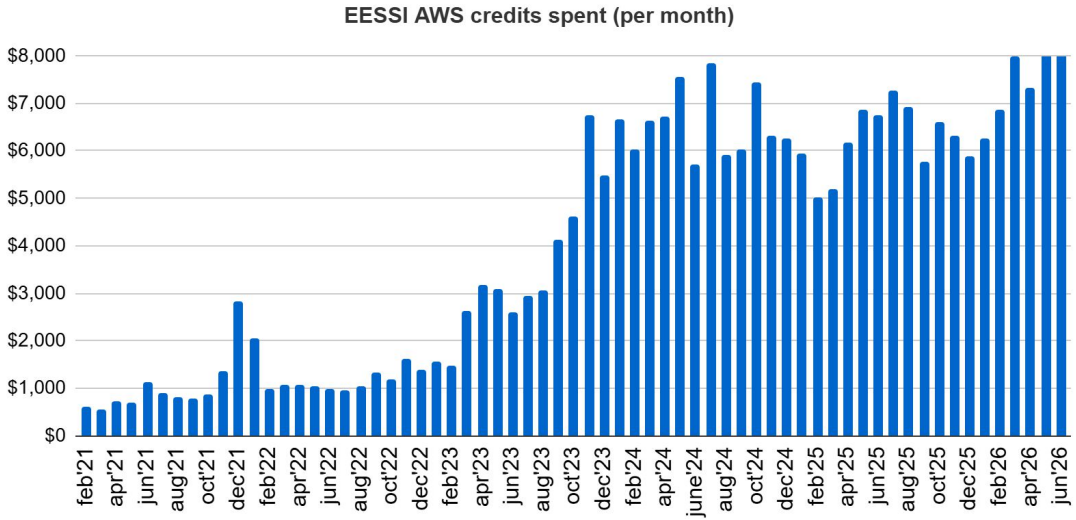
Documentation

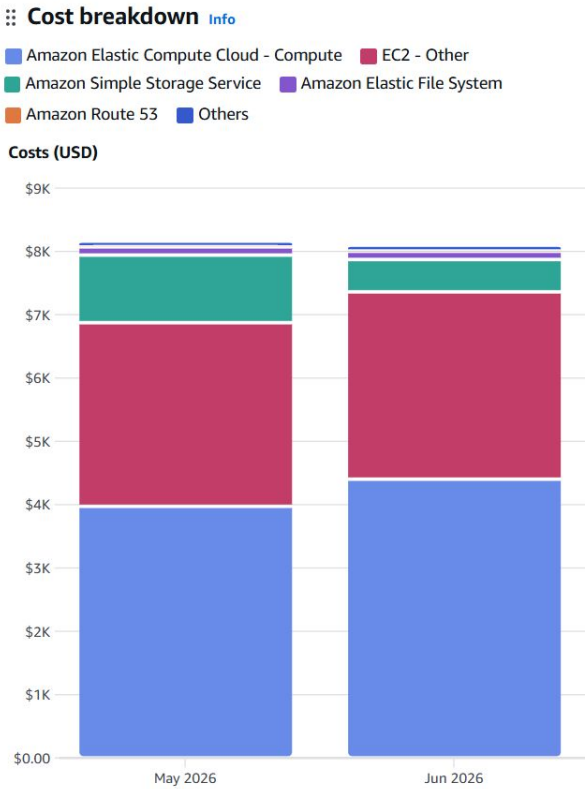
- Document approaches for building on top of EESSI (PR #778)

[Caspar, Bob]

Sponsored AWS credits

- ~$8,162 + $8,098 “spent” in May'26 + Jun'26 on Stratum-1 servers, monitoring, demos, sources.easybuild.io, Slurm clusters (build bot)
- **Bulk of consumed credits due to EESSI build-and-deploy bots**
- **Setting up new build cluster in Ireland (eu-west-1) on hold**
- About 100k credits for 2026 left 🎉
- Concerns about future AWS credits





EESSI Steering Committee

- Most recent ESC meeting was held on 12 May 2026, next one planned for Tue 4 Aug 2026
- Agenda items of last ESC meeting:
  - Short-term roadmap for EESSI => published at https://www.eessi.io/docs/roadmap/
  - Joining HPSF
  - EESSI 2026.x
  - Briefly: EuroHPC funding, making meeting notes public, leftover sponsoring

[Kenneth]

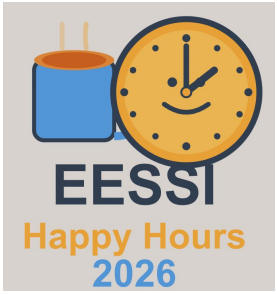
EESSI Happy Hour sessions ⏰ **Mondays 14:00-15:00** CE(S)T

EESSI Happy hour sessions 2026:

- 18 sessions in 2026
  https://www.eessi.io/docs/training-events/happy-hours-sessions/

- Slides and video recordings of all earlier sessions
  https://www.eessi.io/docs/training-events/2026/happy-hours-previous-sessions/

- 🔔 Monday, 29 June 2026 was the final EESSI Happy hour session before the summer break.



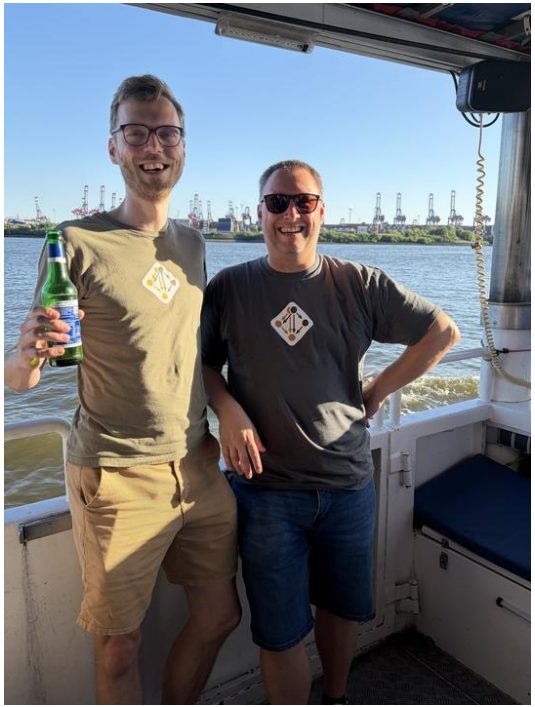
https://eessi.io/docs/training-events/happy-hours-sessions/

[Richard]

ISC'26 conference

- https://isc-hpc.com, 22-26 June 2026 @ Hamburg, Germany

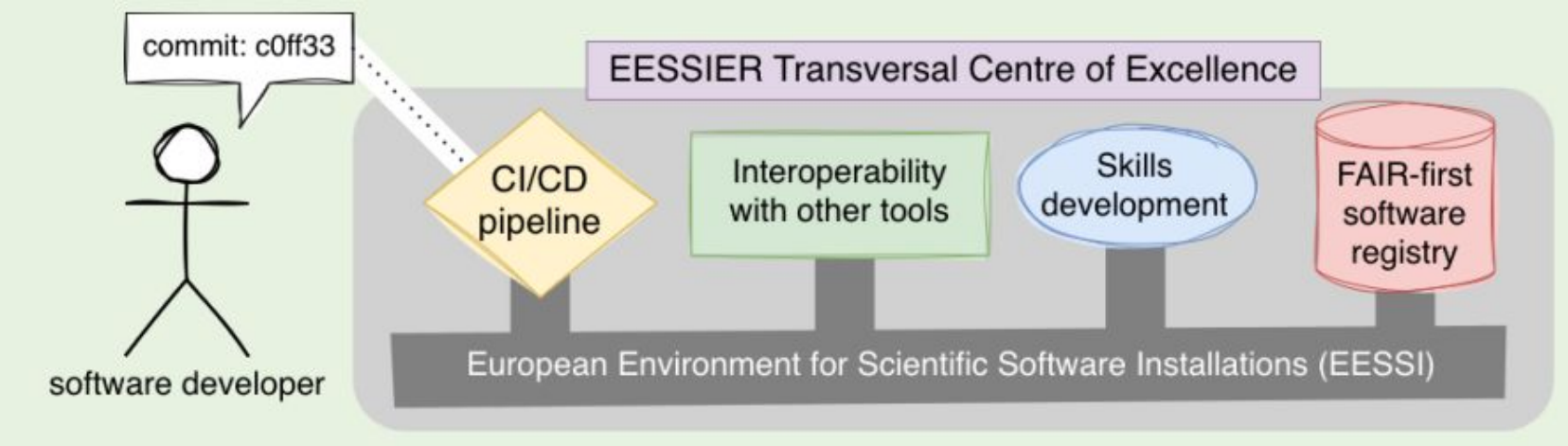
- Half-day tutorial on EESSI (Mon 22 June 2026)
  - About 20 participants
  - Material at eessi.io/isc26-tutorial
- Birds-of-a-Feather sessions on both EESSI & EuroHPC Federation Platform (EFP)
  - EFP: Wed 24 June | EESSI: Thu 25 June
  - Large numbers of attendees, lots of discussion
- We are everywhere! Lot's of potential collaborations and discussions.
  - Blog post(s) in the works



[Kenneth]

EESSIER

- EESSI a critical component of a Transversal EuroHPC CoE
- Project to start in early 2027
- More details to come…for now we just share the good news



EESSI mailing list eessi@list.rug.nl

- Low-frequency mailing list
  - Announcing bi-monthly meetings
  - Important updates and announcements
- To (un)subscribe, send an email to listserv@list.rug.nl with the following body:
  - Subscribe:

    ```
    SUBSCRIBE EESSI Firstname Lastname
    ```

  - Unsubscribe:

    ```
    UNSUBSCRIBE EESSI
    ```

Next EESSI update meetings

- EESSI update meetings are **bi-monthly** (odd months)
- First Thursday of the month at 14:00 CE(S)T, only in odd months
  - iCalendar URL for calendar integration
- Next meetings:
  - Thu 3 Sept 2026 14:00 CEST (12:00 UTC)
  - Thu 5 Nov 2026 14:00 CET (13:00 UTC)
  - Thu 7 Jan 2026 14:00 CET (13:00 UTC)